



Calculation of times to exhaustion at 100 and 120% maximal aerobic speed

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The aim was to compare physiologic responses during exhaustive runs performed on a treadmill at 100 and 120% maximal aerobic speed (MAS: the minimum speed that elicits $\dot{V}O_{2max}$). Fourteen subelite male runners (mean $\pm$ SD; age = 27 ± 5 years; $\dot{V}O_{2max} = 68.9 \pm 4.6$ ml.kg⁻¹.min⁻¹; MAS = 21.5 ± 1 km.h⁻¹) participated. Mean time to exhaustion ($t_{lim100\%}$) at 100% MAS (269 ± 77 s) was similar to those reported in other studies. However, there was large variability in individual $t_{lim100\%}$ MAS (CV = 29%). MAS was positively correlated with $\dot{V}O_{2max}$ ($r = 0.66$, $p < 0.05$) but not with $t_{lim100\%}$ MAS ($r = -0.50$, $p < 0.05$). $t_{lim100\%}$ MAS was correlated with t_{lim} at 120% MAS ($r = 0.52$, $p < 0.05$) and to blood pH following the rest at 120% MAS ($r = -0.68$, $p < 0.05$). The data suggest that running time to exhaustion at MAS in subelite male runners is related to time limit at 120% ($t_{lim120\%}$) MAS. Moreover, anaerobic capacity determined by the exercise to exhaustion at 120% MAS can be defined as the variable 'a' in the model of Monod and Scherrer (1954).

1. Introduction

Maximal aerobic speed (MAS) can be defined as the minimal speed that elicits the maximal oxygen consumption (Di Prampero *et al.* 1986, Lacour *et al.* 1991). MAS has been determined in several studies (Briggs 1977, Peronnet *et al.* 1987, Camus *et al.* 1988), and the times to exhaustion (t_{lim}) at different percentages of MAS have also been investigated (Davies and Thompson 1979, Conley and Krahenbuhl 1980, Leger *et al.* 1986). Recently, mean t_{lim} at MAS was found to be reproducible at a 1 week interval (Billat *et al.* 1994a). Nevertheless, the individual coefficient of variation (CV) of t_{lim} was considerable, $\sim 25\%$, whereas $\dot{V}O_{2max}$ was more consistent (CV of %) (Billat *et al.* 1994a). Moreover, an inverse relationship between $\dot{V}O_{2max}$ and t_{lim} at MAS has been shown (Billat *et al.* 1994b).

The aim of this study was to investigate the intersubject variability in t_{lim} at MAS in a group of individuals that was homogeneous with regard to $\dot{V}O_{2max}$. It was hypothesized that large intersubject variability in t_{lim} at MAS would be due to difference in anaerobic capacity. According to Scott *et al.* (1991), it seems that a

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running time at 120% MAS elicited anaerobic capacity. Medbo *et al.* (1988) suggested that O_2 deficit or anaerobic capacity could be determined during an exhausting exercise of < 5 min. This exercise is not statistically different to the accumulated O_2 deficit between exhausting bouts lasting 2–4 min. To that end, responses at 100 and 120% MAS were compared, with anaerobic capacity determined from data collected during exhaustive exercise at 120% MAS. Moreover, to understand better the great variability of $t_{lim100\%}$ in a homogenous group of $\dot{V}O_{2max}$, this study directly examined the relationship of $t_{lim100\%}$ with $t_{lim120\%}$.

2. Material and method

2.1. Subjects

Fourteen subelite male runners participated in the study. Their mean ($\pm$ SD) age, height and weight were 27 ± 5 years, 178.2 ± 4.9 cm and 70.2 ± 6.0 kg respectively. All subjects were informed of the possible risks and discomforts involved in the participation in the study, and all provided voluntary written informed consent.

2.2. Protocol

2.2.1. Determination of MAS and $\dot{V}O_{2max}$: Each subject performed three exercise tests on a treadmill (Gymrol 1800, Techmachine, Saint Etienne, France). All tests were performed within 3 weeks. During this period, subjects' training programmes were maintained. The tests were separated by a minimum of 7 days, and subjects abstained from high intensity exercise for a least 24 h before each test. In the first test, a continuous incremental protocol was used for the determination of MAS and $\dot{V}O_{2max}$. The initial speed was ~ 12 km.h $^{-1}$ (slope 0%). Speed was increased by 2 km.h $^{-1}$ every 3 min up to the subject's running speed in a 3 km race and then by 1 km.h $^{-1}$ every 3 min thereafter until to exhaustion. During the tests, the subjects breathed through a valve that directed the expired air to a mixing chamber. $\dot{V}O_{2max}$, $\dot{V}E$ and $\dot{V}CO_2$ were computed every 15 s by an automated system (Jaegger Eosprint, Würzburg, Germany). Heart rate (HR) was monitored (Siemens electrocardioscope) and recorded (Sport Tester PE 3000) throughout each test. In the last 30 s of each workload a fingertip blood sample was obtained and analyzed for lactate concentration [L] (Bioblock).

The criteria used to determine $\dot{V}O_{2max}$ were: a plateau in $\dot{V}O_{2max}$ despite an increase in the running speed; a respiratory exchange ratio > 1.1 and HR $> 90\%$ of predicted maximal HR. MAS was defined as the lowest running velocity to elicit a $\dot{V}O_2$ value equal to $\dot{V}O_{2max}$.

2.2.2. Time to exhaustion at 100 and 120% MAS ($t_{lim100\%}$ and $t_{lim120\%}$): Two exhaustive constant-velocity tests were performed. The first was at MAS with a 0% slope. Following a 15-min warm up at 60% MAS, the treadmill speed was quickly increased (within 15 s) to MAS. The subject was then encouraged verbally to run to exhaustion. For the second test, at 120% MAS the slope was constant at 1.5%, as recommended by Margaria *et al.* (1965). In each test, time to exhaustion was measured to the nearest 1.0 s from the moment that MAS (or 120% MAS) was achieved until the subject grasped the handrail and the treadmill was stopped. The same cardiorespiratory measures obtained during the incremental tests were obtained during the constant velocity tests. In addition, arterial blood samples were

obtained from the subject's earlobe before and immediately after each constant velocity test. These samples were analysed using an AVL 993 to determine blood pH.

2.2.3. Statistics: Descriptive data are presented as means $\pm$ SD. Means were compared using Student's *t*-test for dependent means. Correlations between bioenergetic measures were determined. For all analyses, the significant level, *p*, was set at 0.05.

3. Results

Table 1 shows the maximal oxygen uptake of subjects (and CV) averaged $68.9 \pm 4.6 \text{ ml.kg}^{-1}.\text{min}^{-1}$ (7%), ranging from 61.7 to 75 $\text{ml.kg}^{-1}.\text{min}^{-1}$. The average velocity of MAS was $21.5 \pm 1 \text{ km.h}^{-1}$ (4%). Mean times to exhaustion at 100 and 120% MAS were $269 \pm 77 \text{ s}$ ($269 \pm 77 \text{ s}$, CV = 29%) and $86 \pm 25 \text{ s}$ ($86 \pm 25 \text{ s}$, CV = 29%) respectively. $t_{\text{lim}100\%}$ and 120% MAS were positively correlated ($r = 0.52$) (figure 1). Table 2 shows that $\dot{V}\text{O}_{2\text{max}}$ were positively correlated with MAS ($r = 0.66$), and that $t_{\text{lim}100\%}$ and 120% MAS were not significantly correlated with MAS.

Mean pH before performance of the test at 100% and 120% MAS was 7.42 ± 0.01 and 7.42 ± 0.02 . At the end of these two tests, pH was 7.26 ± 0.05 and 7.29 ± 0.06 respectively, and it was not significantly different ($t = 2.13$). In tests at both 100 and 120% MAS, t_{lim} was negatively correlated with blood pH ($r = -0.58$ and -0.69 respectively) (figure 2.). Mean blood [L] after exhaustive exercise at 100 and 120% MAS was 6.8 ± 1.9 and $7 \pm 2.2 \text{ mm}$ respectively. $t_{\text{lim}100\%}$ and 120% MAS were not significantly correlated with blood [L] ($r = 0.51$ and 0.47 respectively). After tests at 100% MAS, pH and [L] were strongly correlated ($r = -0.94$): the relationship was not noted after exercise at 120% MAS ($r = -0.46$).

Table 3 shows the means for cardiovascular variables obtained during exercise at 100 and 120% MAS. They represent 15-s means which, in all cases, were obtained in the final minute of exercise. Results of *t*-tests revealed that there were no significant differences in maximal $\dot{V}\text{O}_{2\text{max}}$, HR and $\dot{V}\text{E}$ obtained in the tests at 100 and 120% MAS. A significant difference was found only for $\dot{V}\text{CO}_{2\text{max}}$. There were no differences between $\dot{V}\text{O}_{2\text{max}}$ measured during the triangular exercise and those reported during the two tests of t_{lim} .

4. Discussion

The $t_{\text{lim}100\%}$ MAS reported in this study are similar to those reported previously (McLellan and Skinner 1985, Lavoie and Mercier 1987, McLellan and Cheung 1992, Billat *et al.* 1994a). The relation between $t_{\text{lim}100\%}$ MAS and MAS is negatively significant ($r = -0.50$). These data confirm that running time to exhaustion at MAS in subelite distance runners is inversely correlated with $\dot{V}\text{O}_{2\text{max}}$, in accordance with the results of Billat *et al.* (1994b). Subjects capable of sustaining their MAS for a longer period are those who have a lower $\dot{V}\text{O}_{2\text{max}}$. It is concluded that truly maximal $\dot{V}\text{O}_{2\text{max}}$ were obtained since there were no differences between the results of incremental and two constant power tests.

These results were more important because the tests of exercise to exhaustion depend on the good evaluation of $\dot{V}\text{O}_{2\text{max}}$. The mean pH at the end of the exhaustive runs at 100 and 120% MAS was not significantly different. The mean [L] was also not significantly different, showing the importance of the anaerobic capacity during exercise at MAS.

Table 1. Individual data

Subjects	$\dot{V}O_{2max}$ (ml.min.kg ⁻¹)	MAS (km.h ⁻¹)	t_{lim100} (s)	t_{lim120} (s)	pH rest (t_{lim100})	pH end (t_{lim100})	pH rest (t_{lim120})	pH end (t_{lim120})	[L] (mM) (t_{lim100})	[L] (mM) (t_{lim120})
1	71.6	21.0	289	95	7.41	7.27	7.40	7.26	6.6	2.6
2	71.1	22.0	189	60	7.40	7.28	7.43	7.41	5.8	7.2
3	72.2	22.5	283	70	7.43	7.27	7.43	7.29	5.7	6.4
4	69.6	22.5	227	74	7.41	7.25	7.41	7.31	7.5	5.8
5	61.7	21.0	367	76	7.44	7.24	7.38	7.24	7.5	7.7
6	64.5	21.0	332	85	—	—	7.41	7.27	8.6	7.9
7	75.0	23.5	158	61	7.43	7.26	7.44	7.27	7.2	9.0
8	67.9	22.0	192	120	7.42	7.26	7.42	7.24	8.3	8.9
9	63.4	20.0	203	70	7.41	7.34	7.41	7.32	3.1	4.6
10	68.6	21.5	372	131	7.39	7.15	7.42	7.21	9.9	11.1
11	68.5	21.5	300	74	7.42	7.31	7.43	7.38	3.9	5.5
12	74.9	21.5	171	63	7.43	7.31	7.42	7.36	5.3	5.9
13	61.7	20.0	352	134	7.43	7.26	7.42	7.25	7.7	8.7
14	73.5	21.0	337	86	7.41	7.22	—	—	8.7	6.1
Mean	68.9	21.5	269	86	7.42	7.26	7.42	7.29	6.8	7.0
SD	4.6	1	77	25	0.01	0.05	0.02	0.06	1.1	2.2
CV%	7	4	29	29	1	7	3	8	16	31

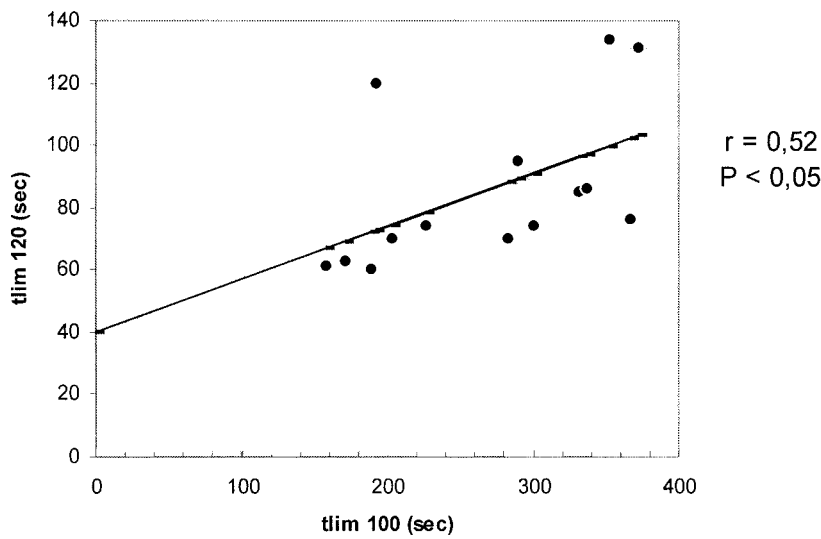


Figure 1. The relationship between tlim at MAS and tlim at 120% of MAS.

Table 2. Correlation between MAS and the other characteristics.

	<i>r</i>
$\dot{V}O_{2m\ ax}/MAS$	0.66
$t_{lim\ 100}/MAS$	− 0.50
$t_{lim\ 120}/MAS$	− 0.39

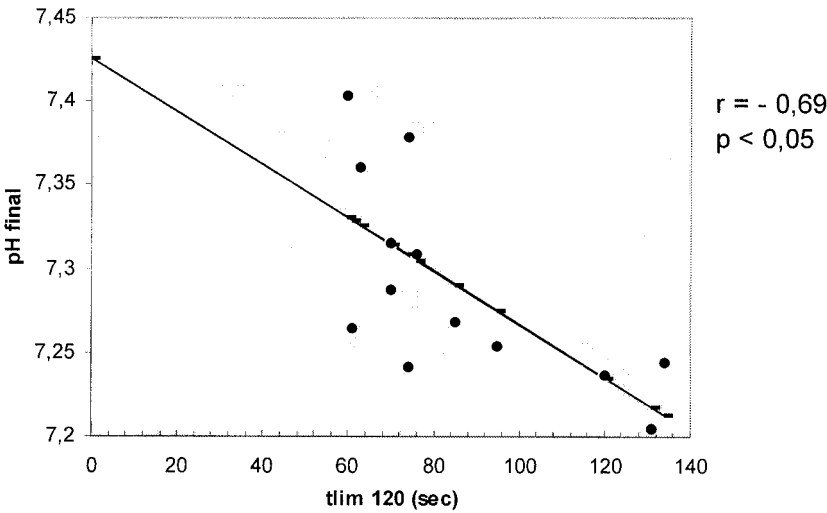


Figure 2. The relationship between tlim at 120% of MAS and blood pH at the end.

Table 3. Mean end cardiorespiratory and metabolic values.

	$\dot{V}O_2$ (ml.min ⁻¹ .kg ⁻¹)	$\dot{V}CO_2$ (l.min ⁻¹)	HR (beats.min ⁻¹)	$\dot{V}E$ (l.min ⁻¹)
t_{lim} 100				
Mean	67.0	5.23	188	158.6
SD	6.7	0.54	6	17.7
t_{lim} 120				
Mean	67.5	6.57	186	165.6
SD	6	0.80	7	21.3
<i>t</i> -test	0.25	3.08	1.05	0.81

Moreover, pH and blood [L] confirm that running to exhaustion at 120% MAS is suitable for the determination of the anaerobic capacity as suggested by Scott *et al.* (1991). These findings, despite the lower mean [L] (6.8 and 7.0 mm for $t_{lim100\%}$ and $t_{lim120\%}$ respectively) could be explained by the fact that the group was composed of only long-distance runners.

Mean t_{lim} at 120% MAS was lower than those of Robergs *et al.* (1991), who reported values of 2 min. The values were more important than ours because the determination of $\dot{V}O_{2max}$ was not specified, and if the evaluation of $\dot{V}O_{2max}$ was not correctly realised, MAS can be underestimated. The time to exhaustion at 120% MAS according to Scott *et al.* (1991) and Medbo *et al.* (1988) can represent anaerobic capacity. The results show that there was a significant positive correlation between t_{lim} at 100% and 120% MAS, showing that anaerobic capacity plays a role in the ability to sustain running at MAS, as previously suggested by Billat *et al.* (1994b). Previously, we have shown that t_{lim} at 100% MAS was correlated with t_{lim} at 105% MAS (Billat *et al.* 1994c). If one takes reference on the model explicated by Monod and Scherrer (1954) and Billat *et al.* (1994b):

$$P = \frac{T_{lim}}{t_{lim}}$$

$$T_{lim} = a + bt_{lim} = P.t_{lim}$$

$$t_{lim} = \frac{a}{P - b},$$

where P is power output at $\dot{V}O_{2max}$, b is estimated by the fraction of $\dot{V}O_{2max}$ at the lactate threshold, and a is anaerobic capacity. These results suggested that 'a' in the model of Monod and Scherrer (1954) could be defined as the anaerobic running capacity.

References

- BILLAT, V., RENOUX, J. C., PINOTEAU, J., PETIT, B. and KORALSZTEIN, J. P. 1994a, Reproducibility of running time to exhaustion at in subelite runners, *Medicine and Science in Sports and Exercise*, **26**, 254–257.
- BILLAT, V., RENOUX, J. C., PINOTEAU, J., PETIT, B. and KORALSZTEIN, J. P. 1994b, Time to exhaustion at 100% of velocity at $\dot{V}O_{2max}$ and modeling of the time limit/velocity relationship in 38 elite long distance runners, *European Journal of Applied Physiology and Occupational Physiology*, **69**, 271–273.

- BILLAT, V., RENOUX, J. C., PINOTEAU, J., PETIT, B. and KORALSZTEIN, J. P. 1994c, Time to exhaustion at 90, 100, 105% of speed at $\dot{V}O_{2\max}$ and critical speed in elite long-distance runners, *Archives Internationales de Physiologie, de Biochimie et de Biophysique*, **103**, 129–135.
- BRIGGS, C. A. 1977, Maximum aerobic power and endurance as predictor of middle distance running success, *Austria Journal Sports and Medicine*, **9**, 28–31.
- CAMUS, G., JUCHNES, J., THYS, H. and FOSSION, A. 1988, Relation entre le temps limite et la consommation maximale d'oxygène dans la course supramaximale, *Journal of Physiology, Paris*, **83**, 26–30.
- CONLEY, D. L. and KRAHENBUHL, G. S. 1980, Running economy and distance running performance of highly trained athlete, *Medicine and Science in Sports and Exercise*, **12**, 357–360.
- DAVIES, C. T. M. and THOMPSON, M. W. 1979, Aerobic performance of female marathon and male ultramarathon athletes, *European Journal of Applied Physiology*, **41**, 233–245.
- DI PRAMPERO, P. E., ATCHOU, G., BRUCKNER, J. C. and MOI, C. 1986, The energetics of endurance running, *European Journal of Applied Physiology*, **37**, 83–92.
- LACOUR, JR, J.R., PADILLA-MAGUNACELAYA, S., CHATARD, J. C., ARSAC, L. and BARTHELEMY, J. C. 1991, Assessment of running velocity at maximal oxygen uptake, *European Journal of Applied Physiology*, **62**, 77–82.
- LAVOIE, J. M. and MERCIER, T. H. 1987, Incremental and constant load determinations of $\dot{V}O_{2\max}$ and maximal constant load performance time, *Canadian Journal of Applied Physiology*, **12**, 229–232.
- LEGER, L., MERCIER, D. and GAUVIN, L. 1986, The relationship between $\dot{V}O_{2\max}$ and running performance time, in D. M. Landers (ed.), *Sport and Elite Performers: Proceedings of 1984 Olympic Scientific Congress*, vol. 3 (Champaign: Human Kinetics), 113–120.
- MARGARIA, R., MANGILI, F., CUTTICA, F. and CERRETELLI, P. 1965, The kinetics of oxygen consumption at the onset of muscular exercise in man, *Ergonomics*, **8**, 49–54.
- MCLELLAN, T. M. and CHEUNG, S. Y. 1992, A comparative evaluation of the individual anaerobic threshold and the critical power, *Medicine and Science in Sports and Exercise*, **24**, 543–550.
- MCLELLAN, T. M. and SKINNER, J. S. 1985, Submaximal endurance performance related to the ventilatory thresholds, *Canadian Journal of Applied Physiology*, **10**, 81–87.
- MEDBO, J. L., MOHN, A. C. and TABATA, I. 1988, Anaerobic capacity determined by maximal accumulated O_2 deficit, *Journal of Applied Physiology*, **64**, 50–60.
- MONOD, H. and SCHERRER, J. 1954, The work capacity of a synergic muscular group, *Ergonomics*, **8**, 329–338.
- PERONNET, F., THIBAUT, G., RHODES, E. C. and MCKENZIE, D. C. 1987, Correlation between ventilatory threshold and endurance capability in marathon runners, *Medicine and Science in Sports and Exercise*, **6**, 610–615.
- ROBERGS, R. A., PASCOE, D. D. and COSTILL, D. L. 1991, Effects of warm-up on muscle glycogenolysis during intense exercise, *Medicine and Science in Sports and Exercise*, **23**, 1, 37–43.
- SCOTT, C. B., ROBY, F. B., LOHMAN, G. T. G. and BUNT, J. C. 1991, The maximally accumulated oxygen deficit as an indicator of anaerobic capacity, *Medicine and Science in Sports and Exercise*, **23**, 618–624.